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# Unified and Multi-Task Learning in NLP

① Single task learning limitations
② Pretraining and transfer learning
③ Question answering
④ DecaNLP and MQAN
⑤ Anti-curriculum training and generalization
⑥ Summary and Takeaways

# Objectives

- Understand the limitations of **single-task learning** in NLP.
- Explain the role of **pre-training and transfer learning**.
- Describe how diverse NLP tasks can be unified as **question answering**.
- Understand **DecaNLP and MQAN** as multi-task frameworks.
- Analyze concepts of **anti-curriculum training and zero-shot generalization**.

# Single task learning limitations

# Evolution of Machine Learning

- **Machine Learning with Feature Engineering**
  - Traditional **machine learning models** (e.g., SVMs, decision trees) require **manual feature engineering**.
  - Domain experts **handcraft features** based on prior knowledge.
  - **Example:** In NLP, using **TF-IDF** for text representation.

- **Deep Learning for Feature Learning**
  - **Deep neural networks** automatically **learn features** from raw data.
  - Eliminates the need for manual feature engineering.
  - **Example:** CNNs extract hierarchical features from images, RNNs learn sequential patterns in text.

# Evolution of Machine Learning, cont.

- **Deep Architecture Engineering for a Single Task**
  - Custom **model architectures** designed for specific tasks (e.g., Transformers for NLP, ResNets for vision).
  - Requires **heavy tuning** and **task-specific modifications**.
  - **Example:** BERT and GPT models optimized for text generation and understanding.

- **What's Next?**
  - **Generalized AI models** that learn across multiple domains (**multi-modal learning**).
  - **More efficient deep learning** architectures reducing computational cost.
  - **Few-shot and zero-shot learning** for better adaptability.
  - **Self-supervised learning** to reduce labeled data dependency.

# Limits of Single-Task Learning

- **Significant progress** in ML with better datasets, models, and evaluation metrics.
- **Optimized performance** for specific tasks (e.g., NLP, vision, speech).
- **The Problem: Local Optima in Task-Specific Models**
  - Models excel at **specific benchmarks** but **fail to generalize**.
  - When the dataset size is **manageable (<1000 columns)**, can optimize for performance.
  - **Issue:** Improvements are often **incremental and task-bound**.

- **Towards More General AI**
  - Need **continuous learning** instead of training separate models per task.
  - **Single models** should **adapt, retain knowledge, and transfer learning** across domains.
  - Current models typically start from **random initialization** or **partial pretraining**, but lack **lifelong learning**.

# Pretraining and transfer learning

# Pre-Training and Knowledge Sharing in AI

- **The Power of Pre-Training**
  - Pre-training allows models to **learn general features** from large datasets before fine-tuning for specific tasks.
  - **Significantly improves performance** and **reduces labeled data requirements**.
- **Success in Computer Vision: ImageNet + CC**
  - **ImageNet pre-training** revolutionized vision tasks.
  - **Common Crawl (CC)** further enhanced large-scale vision datasets.
  - **Classification was the key blocking task**, enabling feature transfer to detection, segmentation, and other vision applications.
- **Success in NLP: Evolution of Pre-Trained Models**
  - No single **blocking task** in NLP → **multi-task learning** emerged as a necessity.
  - Major advancements in representation learning:
    - **Word2Vec, GloVe** – Learned static word embeddings.
    - **CoVe, ELMo** – Contextual word representations via LSTMs.
    - **BERT, GPT, T5** – Transformer-based models with deep contextual learning.

# Pre-Training and Knowledge Sharing in AI

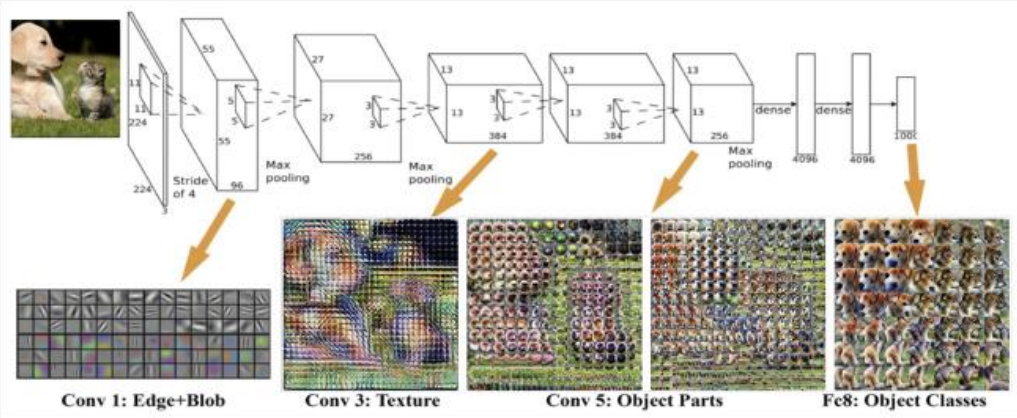
**Key Takeaways**

✅ Pre-training accelerates downstream task performance.

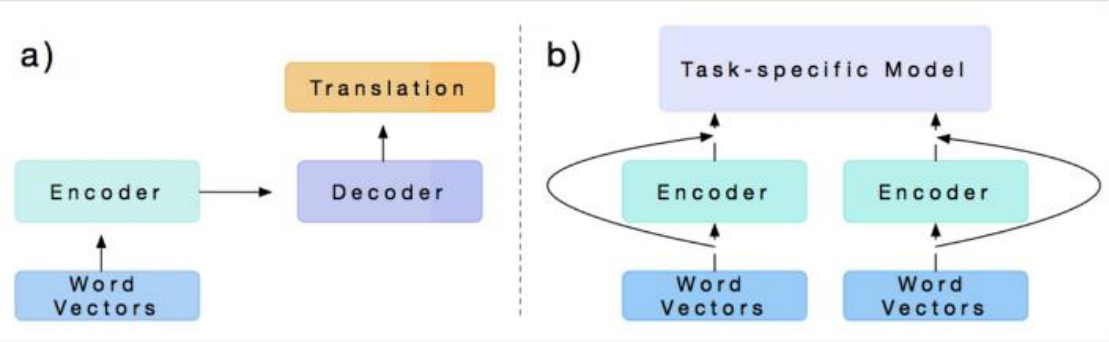
✅ Vision had a clear task (classification), NLP required multi-task learning.

✅ Modern AI relies on large-scale pre-training and transfer learning for efficiency.

**Vision**




**NLP**



# Challenges in NLP-Multiple Reasoning Types

- **Why is NLP Complex?** NLP requires multiple **types of reasoning**:
  - **Logical** – Inferring cause and effect.
  - **Linguistic** – Understanding syntax, semantics, and discourse.
  - **Emotional** – Detecting sentiment and tone.
  - **Visual** – Grounding language in real-world imagery.

- **Short & Long-Term Memory Needs**
  - Understanding a sentence requires **short-term memory** (e.g., resolving pronouns in a paragraph).
  - Comprehending an entire document or conversation needs **long-term memory** (e.g., tracking context across turns).

# Challenges in NLP-Multiple Reasoning Types

- **The Fragmentation of NLP**
  - NLP has been divided into **intermediate and separate tasks** (e.g., parsing, NER, QA) to **enable incremental progress**.
  - Led to **benchmark chasing**, where individual subfields focused on optimizing task-specific models.

- **Can a Single Unsupervised Task Solve NLP?**
  - **No.** Unlike vision, where self-supervised (e.g., contrastive) learning can work well, **language requires supervision**.
  - **Why?** Humans **learn language through structured feedback** (e.g., interaction, correction). Also, understanding meaning **depends on external world knowledge and experience**.

# Task Learning – A Key Challenge

- **Multi-Task Learning as a Blocker for General NLP**
  - NLP requires solving **diverse tasks** (e.g., translation, summarization, reasoning, question answering).
  - Traditional **single-task models** struggle to generalize beyond training objective.
  - **Multi-task learning (MTL)** is essential but remains a major challenge.
- **The Promise of Unified Multi-Task Models**
  - Instead of training separate models for each task, a **single model** can:
    - **Decide how to transfer knowledge** across domains.
    - Use **domain adaptation, weight sharing, transfer learning, and zero-shot learning**.

**Advantages of Unified Multi-Task Models**

- ✅ **Faster adaptation to new tasks** → Eliminates the need for retraining from scratch.
- ✅ **Simplifies deployment** → A single model for multiple tasks reduces production overhead.
- ✅ **Lower barrier for adoption** → Enables more people to apply NLP without task-specific expertise.
- ✅ **Towards Continual Learning** → Models can **learn incrementally**, reducing catastrophic forgetting.

# Expressing Many NLP Tasks in a Framework

- To create a **general NLP model**, different tasks must be expressed in a **consistent framework**. Three common approaches:
- **Sequence Tagging *(Token-Level Tasks):*** Each word in the input is assigned a label.
  - **Named Entity Recognition (NER)** → Identifies entities in text.
  - **Aspect-Specific Sentiment Analysis** → Determines sentiment of specific aspects in a sentence.
- **Text Classification *(Sentence-Level Tasks):*** Assigns a single label to an input text.
  - **Dialogue State Tracking** → Predicts the current state of a conversation (intent detection).
  - **Sentiment Classification** → Determines whether a sentence expresses +, -, or neutral sentiment.
- **Sequence-to-Sequence (Seq2Seq) Models *(Generation Tasks):*** Converts an input sequence into an output sequence.
  - **Machine Translation** → Translates text from one language to another.
  - **Summarization** → Generates a concise summary of a document.
  - **Question Answering** → Produces an answer given a question and context.

**Why is a Unified Framework Important?**

- ✅ Enables **multi-task learning** and **transfer learning**.
- ✅ Reduces **task-specific engineering** and **model duplication**.
- ✅ Moves NLP closer to **generalized, adaptable AI systems**.

# Three Equivalent Tasks in NLP

- **Language Modeling:** The foundational task in NLP where a model **predicts the next word or fills in missing words** in a sentence.
  - **Example:** *"The cat sat on the ___."* → Model predicts *"mat"*.
  - **Why It's Equivalent?** Encodes **world knowledge** and **context**, + for downstream tasks.
- **Question Answering (QA):** Given a context and a question, generate an **explicit answer**.
  - **Example: Context:** *"Marie Curie discovered radioactivity."* **Question:** *"What did Marie Curie discover?"* **Answer:** *"Radioactivity."*
  - **Why It's Equivalent? QA learn contextual reasoning**, how models predict missing info.
- **Dialogue:** Generates **context-aware responses** based on conversational history.
  - **Example: User:** *"Tell me about Marie Curie."* **System:** *"She was a physicist who discovered…"*
  - **Why It's Equivalent? Conversational agents** require both **language modeling** and **question answering skills** to generate meaningful responses.
- Unified models (e.g., GPT, T5) perform well across all three tasks, blurring the distinction.

**Key Insight: All three tasks require similar capabilities**:

✅ Understanding **context** and **world knowledge**.

✅ Handling **long-range dependencies** in text.

✅ Generating **coherent, context-appropriate responses**.

# Three Equivalent Tasks in NLP

| NLP Task | Description | Example | Why It's Equivalent? |
|---|---|---|---|
| **Language Modeling** | Predicts missing or next words in a sentence. | *"The cat sat on the ___."* → *"mat"* | Encodes world knowledge and context, needed for downstream tasks. |
| **Question Answering** | Given a context and a question, generates an explicit answer. | **Context:** *"Marie Curie discovered radioactivity."* **Q:** *"What did Marie Curie discover?"* **A:** *"Radioactivity."* | Requires contextual reasoning, similar to how LMs fill in missing information. |
| **Dialogue** | Generates responses based on conversational history. | **User:** *"Tell me about Marie Curie."* **System:** *"She was a physicist who discovered radioactivity."* | Combines elements of both LM and QA to generate meaningful responses. |

# Multi-Task Learning as Question Answering

- **Unifying NLP Tasks as Question Answering**
  - Many NLP tasks can be **reframed as question-answering problems**, enabling a **single model** to learn across tasks.
  - A unified **QA-based approach** allows for **better generalization** and **knowledge transfer**.

**Advantages of QA-Based Multi-Task Learning**

✅ **Reduces need for task-specific architectures**.

✅ **Enables transfer learning** between related tasks.

✅ **Simplifies model training and deployment**.

✅ **Moves towards more general NLP systems**.

# Multi-Task Learning as Question Answering

| Task | QA Representation |
|---|---|
| Question Answering (QA) | Given a passage and a question, generate an answer. |
| Machine Translation | Question: "How do you say [sentence] in French?" → Answer: "[translated sentence]" |
| Summarization | Question: "What is the summary of this text?" → Answer: "[summary]" |
| Natural Language Inference | Question: "Does sentence A entail sentence B?" → Answer: "Yes/No/Neutral" |
| Sentiment Classification | Question: "What is the sentiment of this review?" → Answer: "Positive/Negative/Neutral" |
| Semantic Role Labeling | Question: "What is the agent of this action?" → Answer: "[subject]" |
| Relation Extraction | Question: "What is the relationship between X and Y?" → Answer: "[relation]" |
| Dialogue | Given conversation history, answer "What should be said next?" |
| Semantic Parsing | Question: "What is the structured meaning of this sentence?" → Answer: "[logical form]" |
| Commonsense Reasoning | Question: "What is the most likely outcome of this event?" → Answer: "[expected result]" |

# Meta-Supervised Learning – Extending Supervision with Task Descriptions

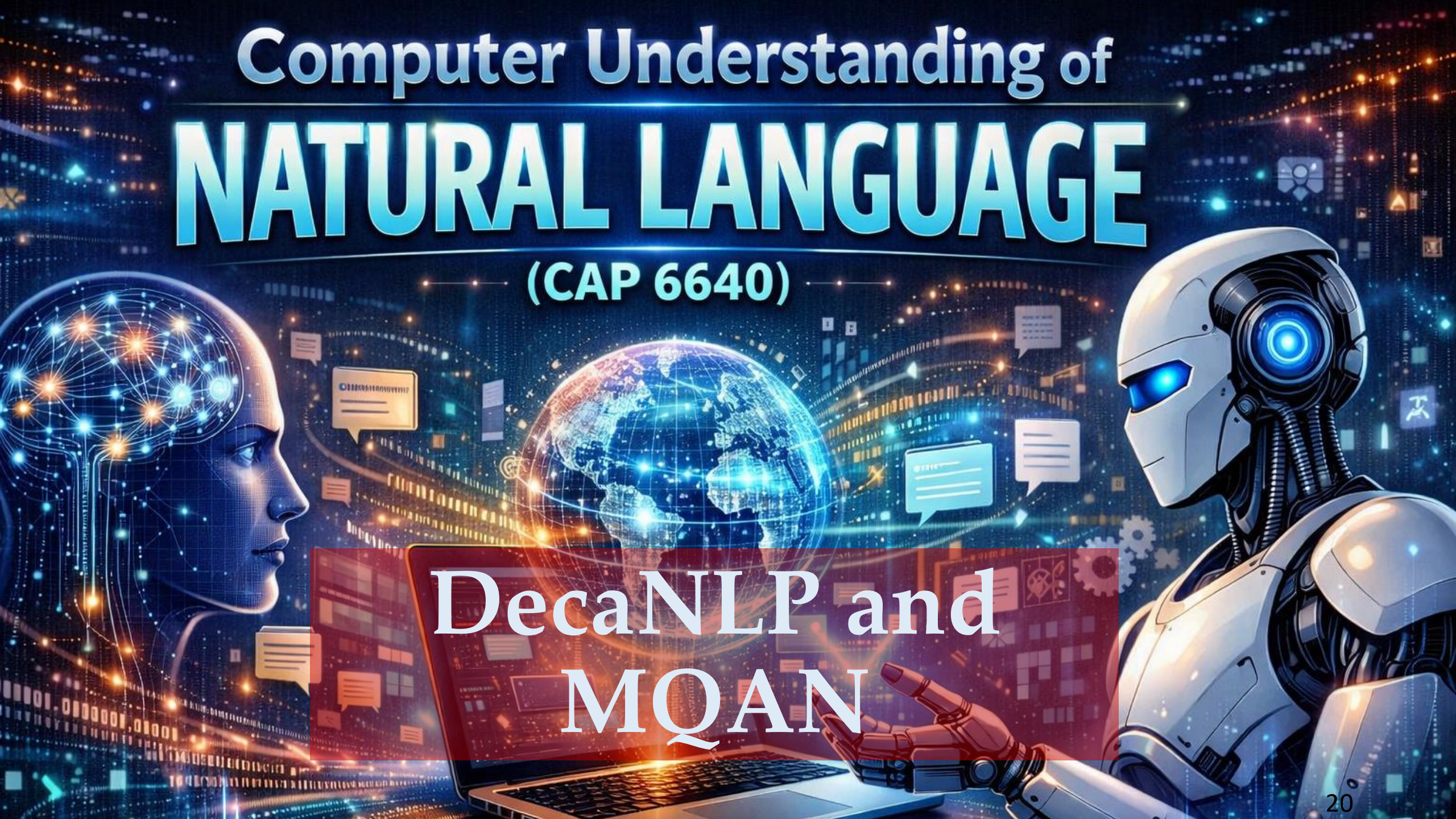
- **What is Meta-Supervised Learning?**
  - Expands traditional supervision from **{x, y}** pairs to **{x, t, y}**, where:
    - **x** = Context (input needed to answer the question).
    - **t** = Task (explicitly defined as a question **q**).
    - **y** = Answer (output based on the task and context).
- **Key Idea: Using Questions as Task Descriptions**
  - Instead of relying only on structured labels, **frame the task as a natural language question (q)**.
  - The model **learns to generalize** by leveraging linguistic information in **q** to connect tasks.

# Meta-Supervised Learning – Extending Supervision with Task Descriptions

| Traditional Supervision | Meta-Supervised Learning |
|---|---|
| Sentiment Classification: "This movie was amazing!" → "Positive" | Task: "What is the sentiment of this review?" Context: "This movie was amazing!" Answer: "Positive" |
| Named Entity Recognition (NER): "Barack Obama was born in Hawaii." → (Obama, PERSON) | Task: "Which words refer to a person?" Context: "Barack Obama was born in Hawaii." Answer: "Barack Obama" |
| Summarization: "The company announced a new product launch." → "New product launch announced." | Task: "What is the summary of this text?" Context: "The company announced a new product launch." Answer: "New product launch announced." |

**Why is This Powerful?**

✅ **Unifies multiple NLP tasks** in a **single framework**.
✅ **Improves transfer learning** by leveraging task descriptions as shared linguistic knowledge.
✅ **Enhances generalization** to unseen tasks with minimal labeled data.




# DecaNLP and MQAN

# DecaNLP – A Unified Approach to NLP Tasks

- **What is DecaNLP?**
  - A **multi-task learning framework** designed to handle diverse NLP tasks **without task-specific modifications**.
  - Models must **internally adjust** to perform different tasks without explicitly knowing the task ID.
- **Key Specifications**
  - **No Task-Specific Modules or Parameters**
    - Unlike traditional multi-task learning, **DecaNLP does not assume access to task IDs**.
    - A **single model architecture** must learn to generalize across all tasks.
  - **Internal Adaptation for Separate Tasks**
    - Instead of using **task-specific heads**, the model must **implicitly recognize the task** based on input structure.
    - The same model should handle **summarization, QA, translation, NER, etc., without retraining**.
  - **Zero-Shot Generalization for Unseen Tasks**
    - **If a model learns from 10 NLP tasks**, it should be able to **extrapolate to new, unseen tasks**.
    - Enables **transfer learning**, where knowledge from one task improves performance on others.

# DecaNLP – A Unified Approach to NLP Tasks

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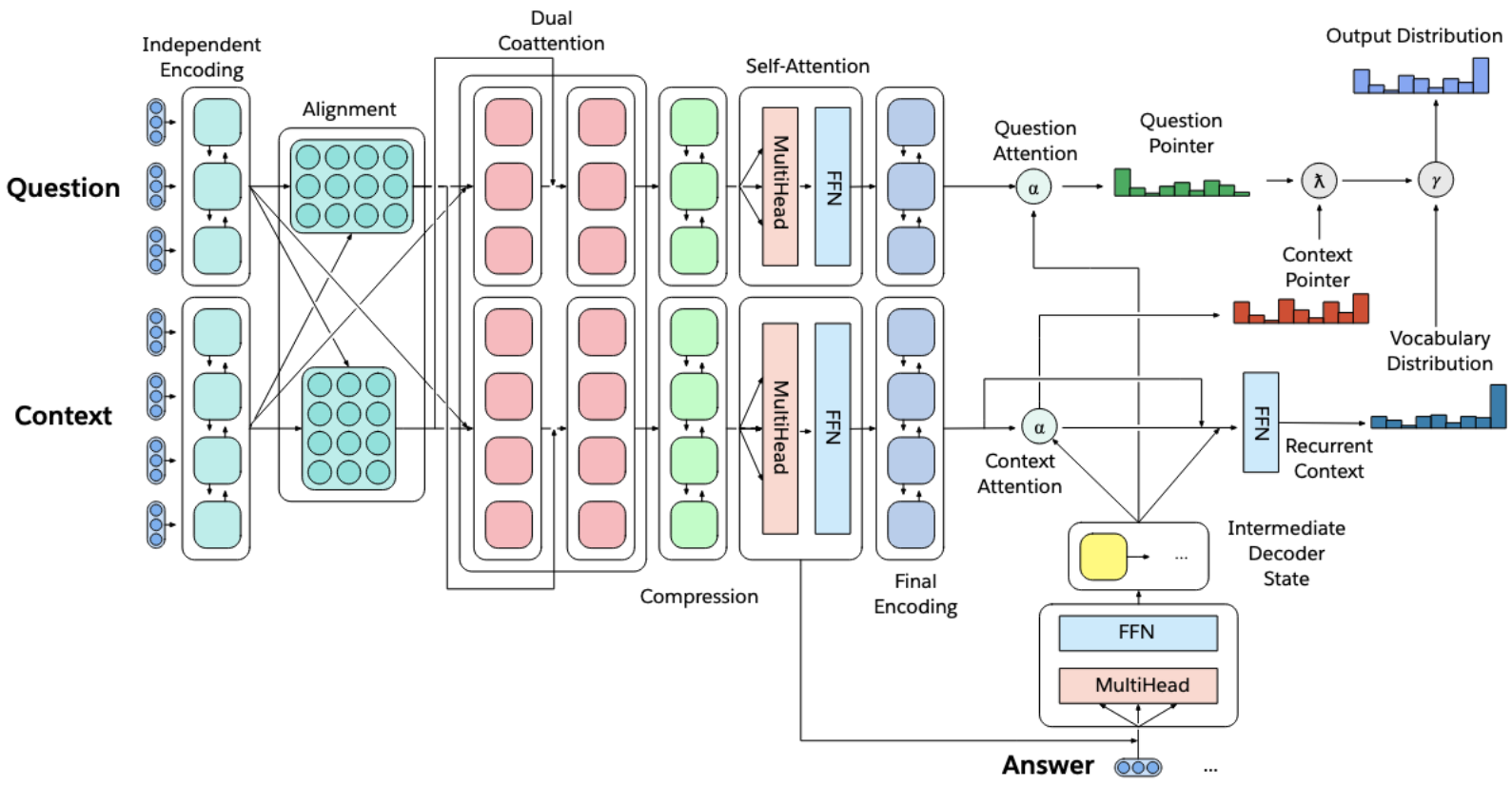
    - Th **raining**.
  - **Zero-S**
    - **If sks**.
    - Er hers.

**Why is This Important?**

- ✅ **Eliminates the need for task-specific architectures.**
- ✅ **Moves towards more generalized NLP models.**
- ✅ **Allows for better adaptation to real-world settings where task boundaries are unclear.**

# DecaNLP – A Unified Approach to NLP Tasks




- **Step 1: Start with Context --** Provide relevant background information.
- **Step 2: Ask a Question --** Frame a question based on the context.
- **Step 3: Generate the Answer Word by Word --** Each word is selected using one of the following methods:
  1. **Point to Context** – Extract words directly from the provided context.
  2. **Point to Question** – Use words from the original question.
  3. **Choose from External Vocabulary** – Select a new word from an external source.
- **Pointer Switch:** At each step, the **Pointer Switch** decides which method to use for generating the next word in the response.

https://arxiv.org/pdf/1806.08730

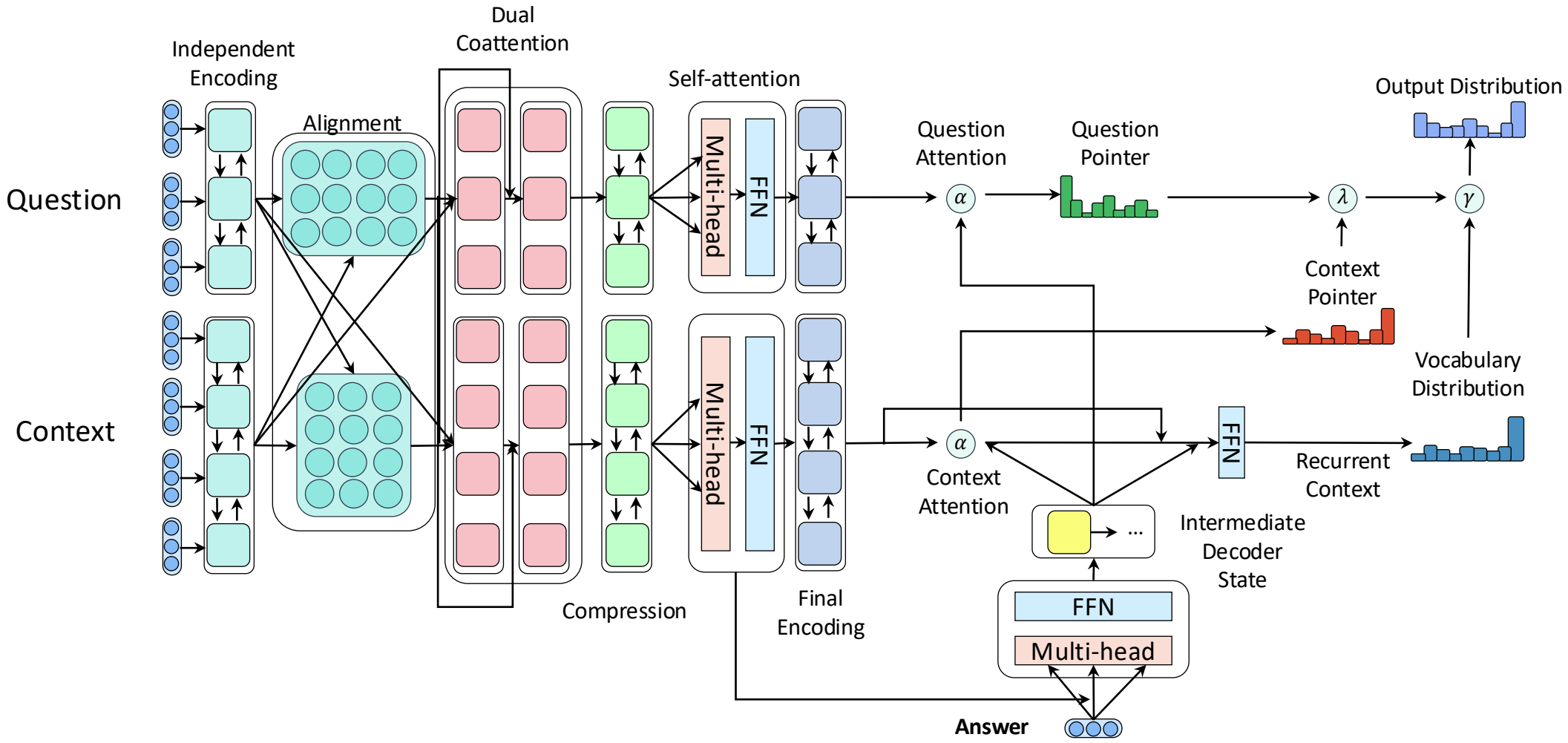
# DecaNLP – A Unified Approach to NLP Tasks

## Examples

| Question | Context | Answer |
|---|---|---|
| What is a major importance of Southern California in relation to California and the US? | ...Southern California is a major economic center for the state of California and the US.... | major economic center |
| What is the translation from English to German? | Most of the planet is ocean water. | Der Großteil der Erde ist Meerwasser |
| What is the summary? | Harry Potter star Daniel Radcliffe gains access to a reported £320 million fortune... | Harry Potter star Daniel Radcliffe gets £320M fortune... |
| Hypothesis: Product and geography are what make cream skimming work. Entailment, neutral, or contradiction? | Premise: Conceptually cream skimming has two basic dimensions – product and geography. | Entailment |
| Is this sentence positive or negative? | A stirring, funny and finally transporting re-imagining of Beauty and the Beast and 1930s horror film. | positive |

| Question | Context | Answer |
|---|---|---|
| What has something experienced? | Areas of the Baltic that have experienced eutrophication. | eutrophication |
| Who is the illustrator of Cycle of the Werewolf? | Cycle of the Werewolf is a short novel by Stephen King, featuring illustrations by comic book artist Bernie Wrightson. | Bernie Wrightson |
| What is the change in dialogue state? | Are there any Eritrean restaurants in town? | food: Eritrean |
| What is the translation from English to SQL? | The table has column names... Tell me what the notes are for South Australia | SELECT notes from table WHERE 'Current Slogan' = 'South Australia' |
| Who had given help? Susan or Joan? | Joan made sure to thank Susan for all the help she had given. | Susan |

# Multitask Q Answering Network (MQAN)



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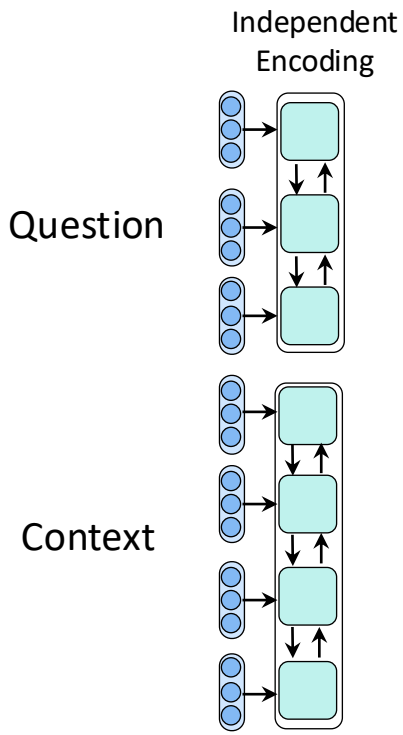
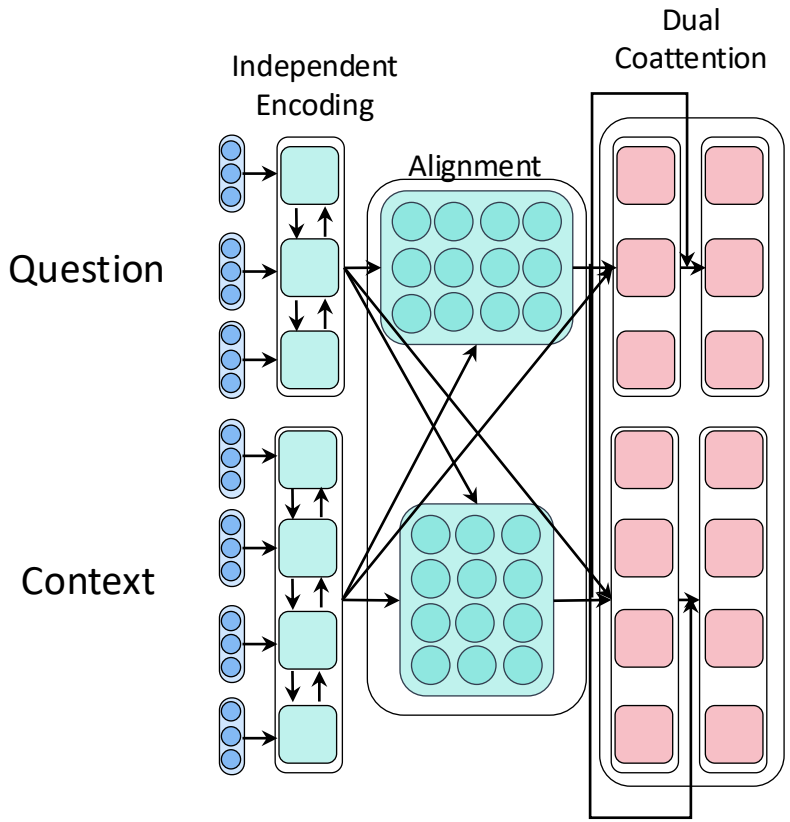


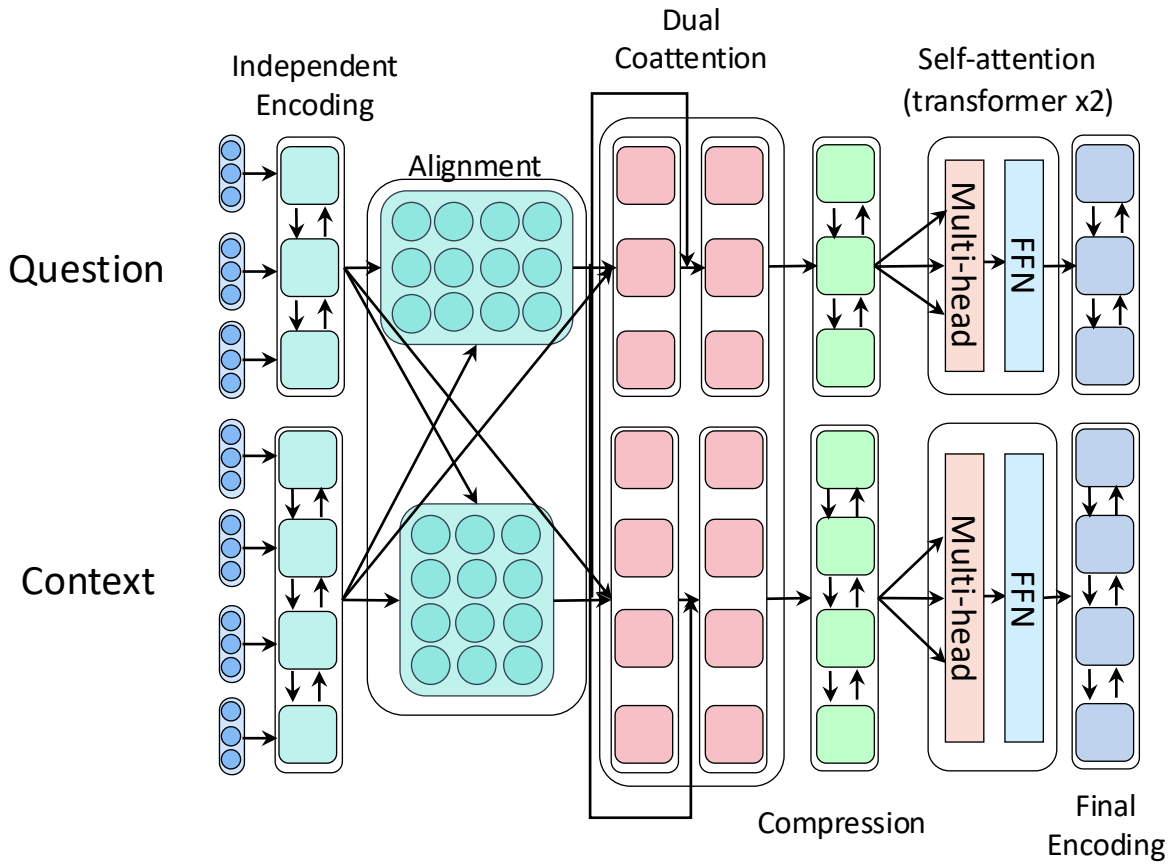

Fixed Glove+Character n-gram embeddings → Linear → Shared BiLSTM with skip connections

# Multitask Q Answering Network (MQAN)



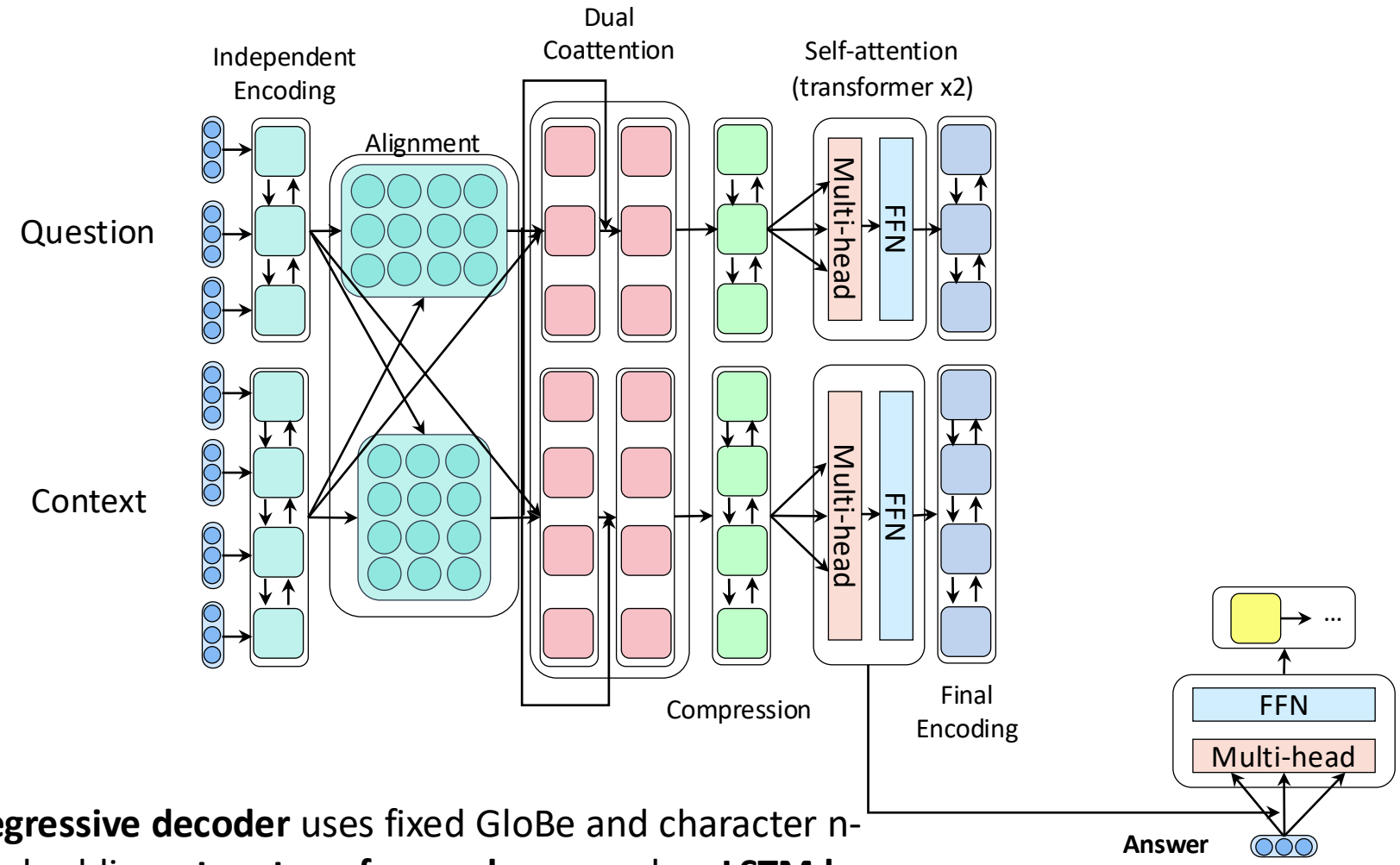

Attention summation from one sequence to the other and back again with skip connections

# Multitask Q Answering Network (MQAN)



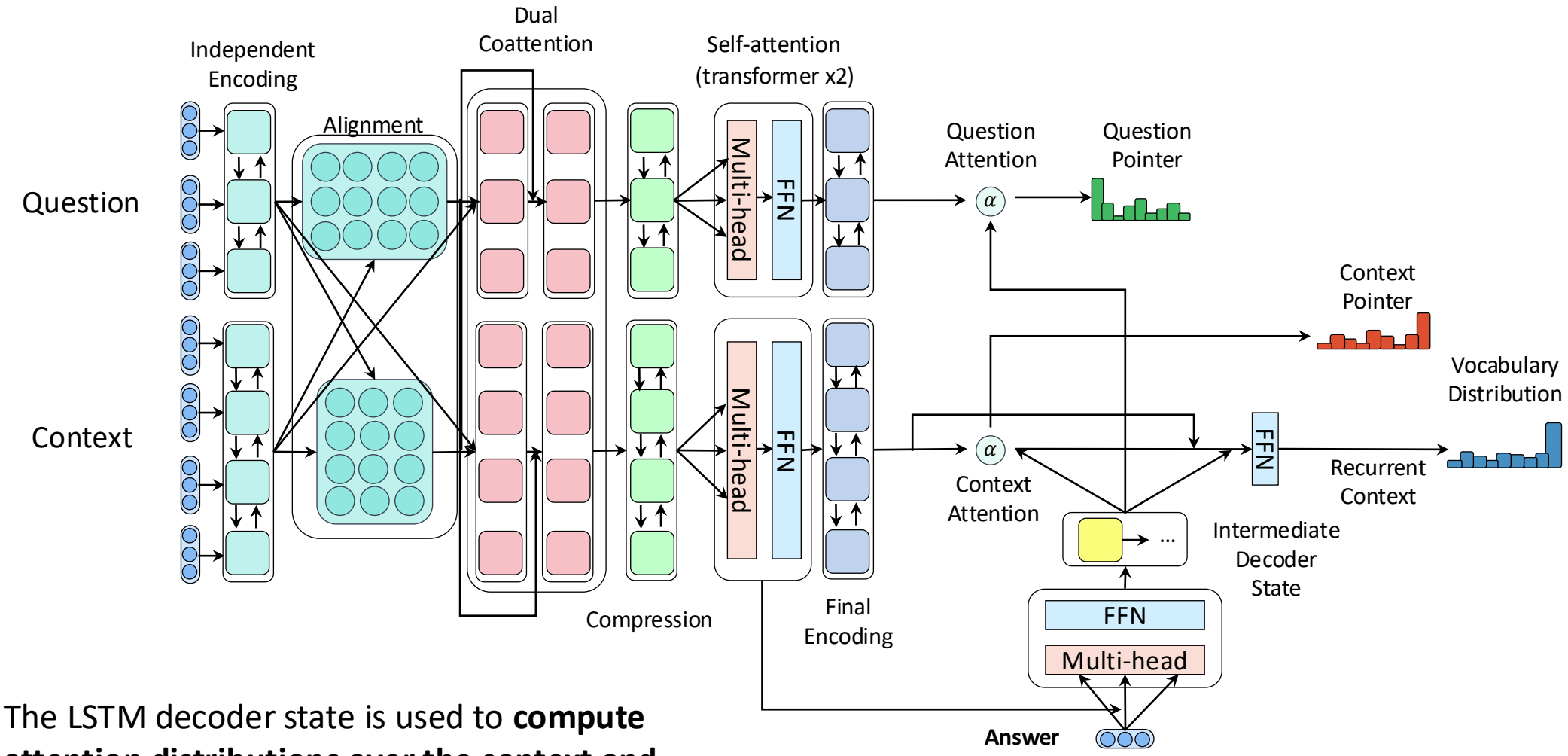

Separate BiLSTMs to reduce dimensionality, two transformer layers, another BiLSTM

# Multitask Q Answering Network (MQAN)



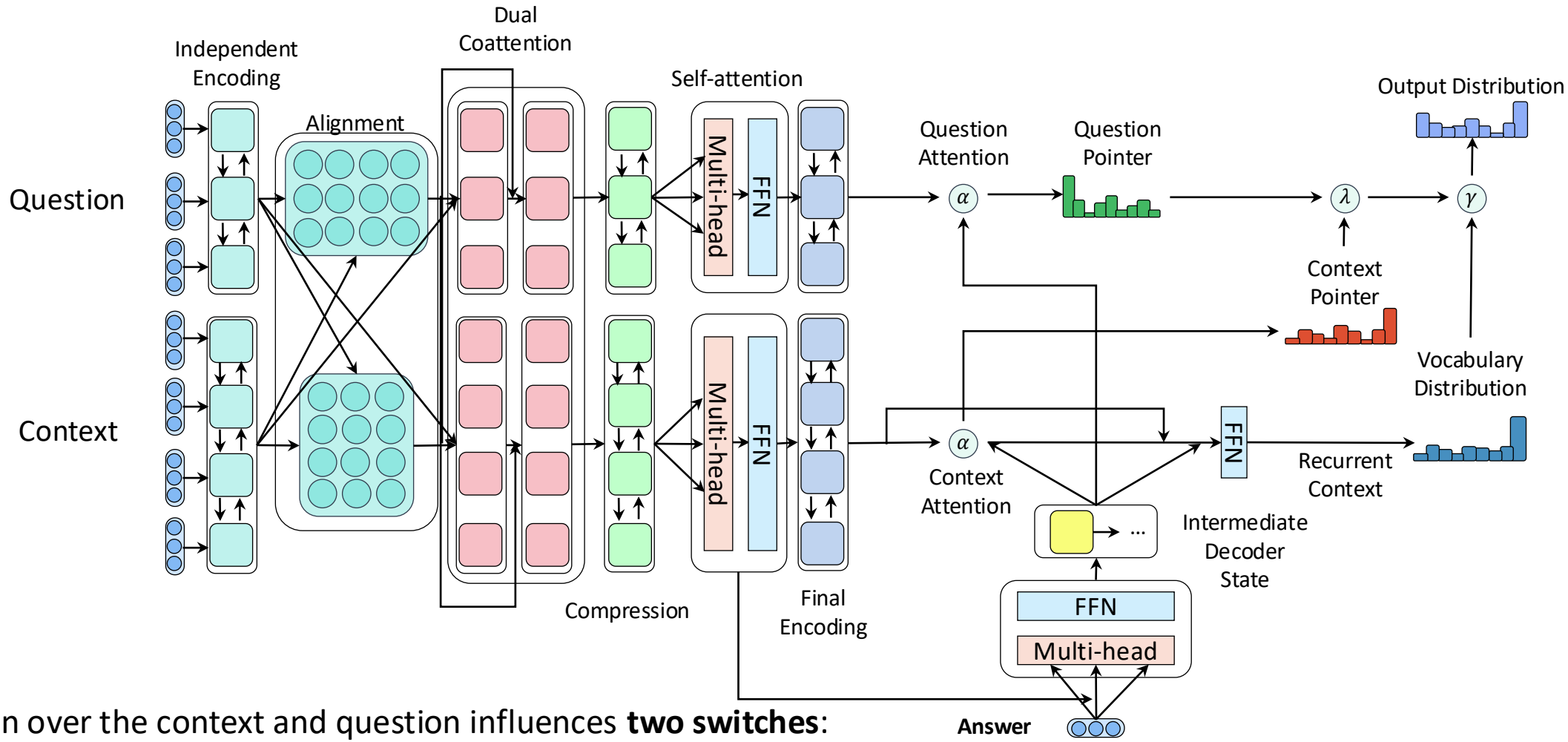

**Auto-regressive decoder** uses fixed GloBe and character n-gram embeddings, **two transformer layers** and an **LSTM layer** that attend to outputs of the last three layers of the encoder

# Multitask Q Answering Network (MQAN)




The LSTM decoder state is used to **compute attention distributions over the context and question** which are used as pointers

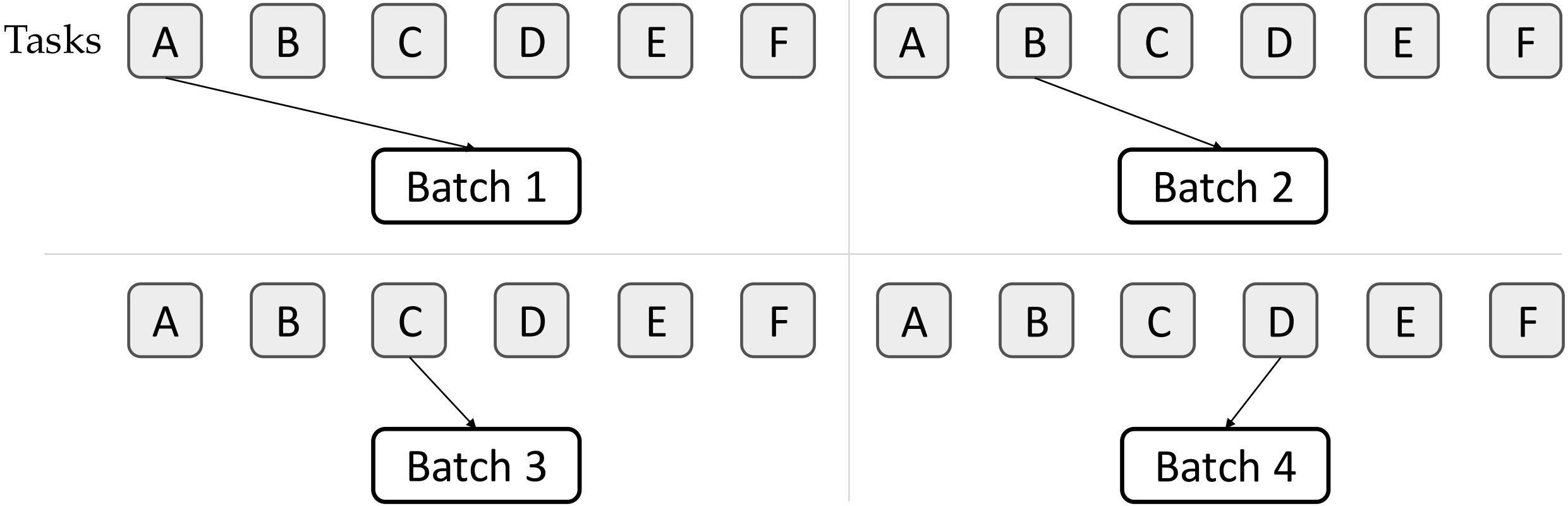
# Multitask Q Answering Network (MQAN)




Attention over the context and question influences **two switches**:
gamma decides whether to copy or select from an **external vocab**,
lambda decides whether to copy from **context or question**

# Anti-curriculum training and generalization

# Fully Joint Training



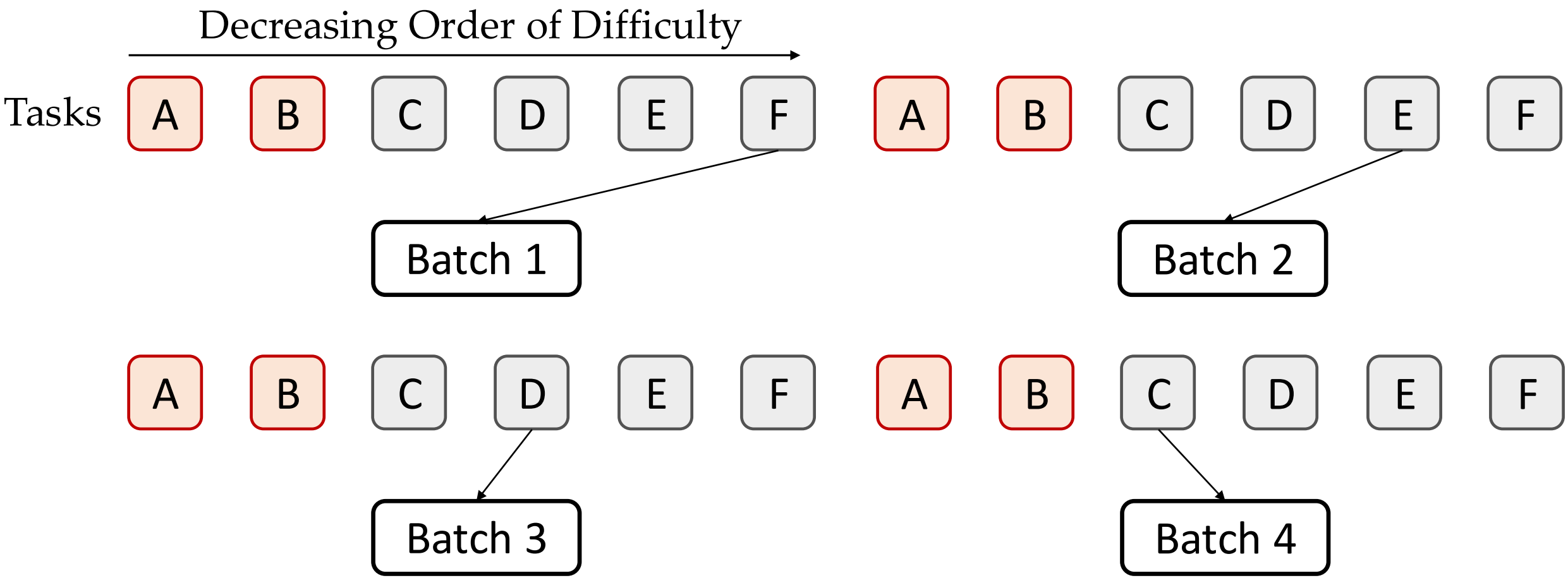
# Anti-Curriculum Pre-Training

- A **pre-training strategy** that **reverses traditional curriculum learning**.
- Instead of **starting with simple tasks** and progressing to harder ones, the model is **first exposed to complex, diverse data**.

- Helps prevent **overfitting to easy patterns** early in training.
  - **Expose the model to a diverse dataset first** (e.g., long-form, ambiguous, multi-domain text).
  - **Gradually introduce simpler, structured tasks later** to refine performance.
  - **Encourage robustness and generalization** by forcing the model to learn meaningful patterns early.
- **Why Use Anti-Curriculum Learning?**
  - ✅ **Improves generalization** – The model doesn't rely on simple heuristics learned early.
  - ✅ **Prevents local minima** – Encourages exploration of complex solutions from the start.
  - ✅ **Boosts performance on diverse, real-world NLP tasks**.

- **Example: NLP Pre-Training Approach**
  - **Curriculum Learning**: Start with basic sentence completion → Progress to reasoning tasks.
  - **Anti-Curriculum Learning**: Start with complex reasoning → Fine-tune on simpler tasks.

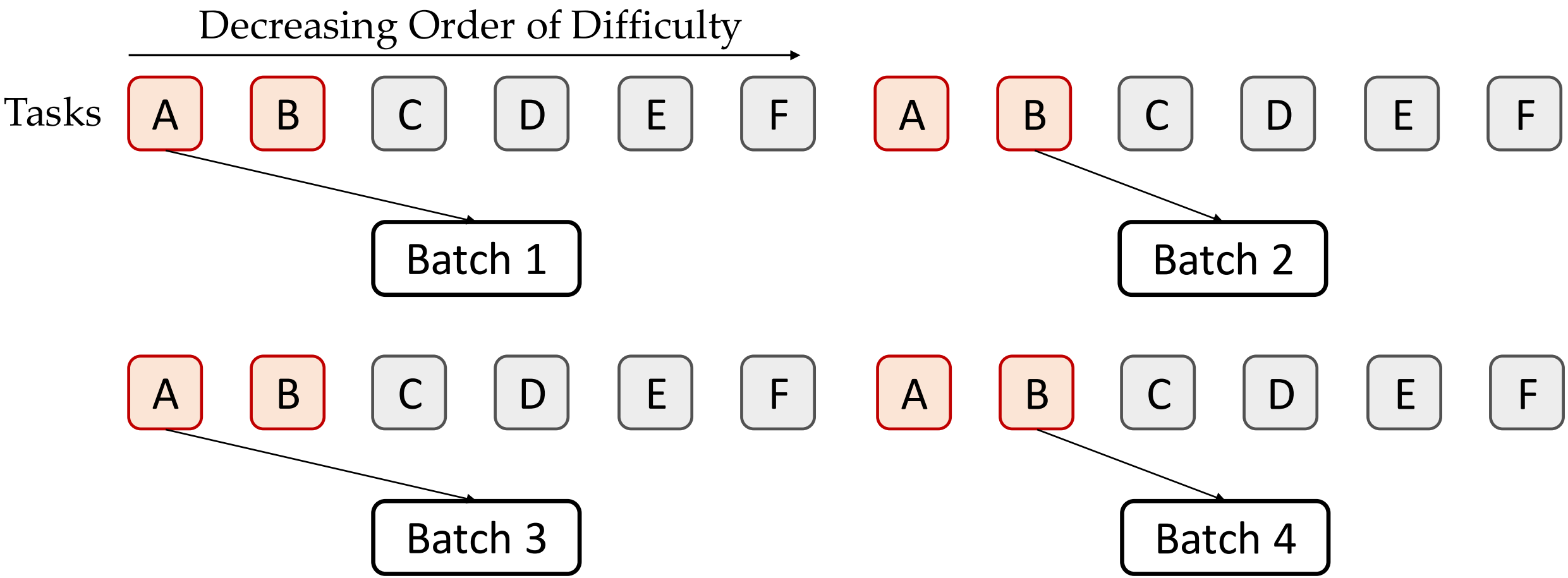
# Curriculum Pre-Training




Difficulty: how many iterations to convergence in the single-task setting

★ Tasks included in the pretraining phase

# Anti-Curriculum Pre-Training




Difficulty: how many iterations to convergence in the single-task setting

★ Tasks included in the pretraining phase

# Zero-Shot Classification, Question Pointers

- **What is Zero-Shot Classification?**
  - A model's ability to **handle new tasks without explicit training**.
  - Uses **question pointers** to reframe labels dynamically, adapting to variations in wording.

- **How Question Pointers Enable Zero-Shot Learning**

  ✅ **Flexible Label Representation** – Allows label variations (e.g., *Positive* → *Happy/Supportive*, *Negative* → *Sad/Unsupportive*).
  ✅ **No Additional Fine-Tuning Required** – The model generalizes across unseen tasks.
  ✅ **Natural Language Understanding** – Converts classification problems into **question-answering** tasks.

# Zero-Shot Classification, Question Pointers

- **Example of Zero-Shot Sentiment Classification**
  - **Context:**
    *C: John had a party but no one came and he was all alone.*
  - **Query Reformulation as QA:**
    *Q: Is this story sad or happy?*
    *A: Sad.*

- **Why This Matters?**
  - Enables models to **perform new tasks** without retraining.
  - Bridges the gap between **classification and QA-based reasoning**.
  - Moves towards **general-purpose NLP models** that require minimal supervision.

# A Unified QA Model for Multiple NLP Tasks

- **Why Frame NLP Tasks as Question Answering?**
  - A single **QA-based model** can generalize across multiple tasks by treating them as different **types of questions**.
  - Enables **more general language understanding** instead of training **task-specific models**.

- **Framework for Tackling Multiple NLP Challenges**

  ✅ **Multi-Task Learning** – Train on multiple NLP tasks using a **single model**.
  ✅ **Domain Adaptation** – Transfer learned knowledge to **new data distributions**.
  ✅ **Transfer Learning** – Leverage **pre-trained language models** for different QA-style tasks.
  ✅ **Weight Sharing & Fine-Tuning** – Train once, adapt to many NLP problems (towards an **ImageNet-CNN equivalent for NLP**).
  ✅ **Zero-Shot Learning** – Handle unseen tasks **without additional training**.

# A Unified QA Model for Multiple Tasks

| NLP Task | QA Reformulation |
|---|---|
| Sentiment Classification | "What is the sentiment of this review?" → "Positive/Negative/Neutral" |
| Named Entity Recognition (NER) | "Which words in this sentence are locations?" |
| Summarization | "What is the summary of this article?" |
| Machine Translation | "How do you say this sentence in French?" |
| Commonsense Reasoning | "If I drop a glass, what happens?" → "It breaks." |

**Key Takeaway**

- ✅ **A single QA-based model** can unify NLP tasks, reducing the need for separate architectures.
- ✅ Moves towards **generalized, adaptable NLP models**, making task deployment **simpler and more scalable**.

# Multitask Q Answering Network (MQAN)

| Task | Dataset | # Train | # Dev | # Test | Metric |
|---|---|---|---|---|---|
| Question Answering | SQuAD | 87599 | 10570 | 9616 | nF1 |
| Machine Translation | IWSLT | 196884 | 993 | 1305 | BLEU |
| Summarization | CNN/DM | 287227 | 13368 | 11490 | ROUGE |
| Natural Language Inference | MNLI | 392702 | 20000 | 20000 | EM |
| Sentiment Analysis | SST | 6920 | 872 | 1821 | EM |
| Semantic Role Labeling | QA-SRL | 6414 | 2183 | 2201 | nF1 |
| Zero-Shot Relation Extraction | QA-ZRE | 840000 | 600 | 12000 | cF1 |
| Goal-Oriented Dialogue | WOZ | 2536 | 830 | 1646 | dsEM |
| Semantic Parsing | WikiSQL | 56355 | 8421 | 15878 | lfEM |
| Pronoun Resolution | MWSC | 80 | 82 | 100 | EM |

Summary of openly available benchmark datasets in decaNLP and evaluation metrics that contribute to the decaScore:

- nF1 is the normalized F1 metric used by SQuAD that strips out articles and punctuation.
- EM is an exact match comparison: for text classification, this amounts to accuracy
- dialogue state exact match (dsEM) is equivalent to exact match of logical forms (lfEM).
- Corpus level F1 (cF1) takes into account that some question are unanswerable.

# Multitask Q Answering Network (MQAN)

| | Single-task Training | | | | Multitask Training | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Dataset | S2S | w/SAtt | +CAtt | +QPtr | S2S | w/SAtt | +CAtt | +QPtr | +ACurr |
| SQuAD | 48.2 | 68.2 | 74.6 | 75.5 | 47.5 | 66.8 | 71.8 | 70.8 | 74.3 |
| IWSLT | 25.0 | 23.3 | 26.0 | 25.5 | 14.2 | 13.6 | 9.0 | 16.1 | 13.7 |
| CNN/DM | 19.0 | 20.0 | 25.1 | 24.0 | 25.7 | 14.0 | 15.7 | 23.9 | 24.6 |
| MNLI | 67.5 | 68.5 | 34.7 | 72.8 | 60.9 | 69.0 | 70.4 | 70.5 | 69.2 |
| SST | 86.4 | 86.8 | 86.2 | 88.1 | 85.9 | 84.7 | 86.5 | 86.2 | 86.4 |
| QA-SRL | 63.5 | 67.8 | 74.8 | 75.2 | 68.7 | 75.1 | 76.1 | 75.8 | 77.6 |
| QA-ZRE | 20.0 | 19.9 | 16.6 | 15.6 | 28.5 | 31.7 | 28.5 | 28.0 | 34.7 |
| WOZ | 85.3 | 86.0 | 86.5 | 84.4 | 84.0 | 82.8 | 75.1 | 80.6 | 84.1 |
| WikiSQL | 60.0 | 72.4 | 72.3 | 72.6 | 45.8 | 64.8 | 62.9 | 62.0 | 58.7 |
| MWSC | 43.9 | 46.3 | 40.4 | 52.4 | 52.4 | 43.9 | 37.8 | 48.8 | 48.4 |
| decaScore | - | - | - | - | 473.6 | 546.4 | 533.8 | 562.7 | 571.7 |

- w/SAtt: with self-attentive transformer layers
- +CAtt: the addition of coattention over a split context and question
- +QPtr: question pointer.
- The last model is equivalent to MQAN.
- Multitask models use a round-robin batch-level sampling strategy to jointly train on the full decaNLP.
- The last column includes an additional anti-curriculum (+ACurr)

# Takeaways

- NLP lacks a single "blocking task"; **multi-task learning is essential**.
- Many tasks (LM, QA, dialogue, classification) are **functionally equivalent**.
- Framing tasks as **QA enables unified modeling and transfer**.
- MQAN demonstrates task-agnostic, pointer-based multi-task learning.
- Unified models move NLP toward **generalized, adaptable AI systems**.